ARTIFICIAL INTELLIGENCE AND GENERATIVE AI

A Beginner’s Study Guide

UNDERGRADUATE MINI PROJECT RESOURCE

AI Personal Study Assistant

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About this guide
A practical introduction to AI and generative AI, with clear explanations, everyday examples, responsible-use guidance, a glossary and practice questions. Its sections are organised to support student revision and question-answering in an AI Personal Study Assistant.

Understand the concepts • Build your skills • Revise with confidence

1. Introduction to Artificial Intelligence

Artificial intelligence (AI) is a field of computing focused on building systems that perform tasks commonly associated with human intelligence. These tasks include recognising patterns, understanding language, making predictions, solving problems, planning and supporting decisions. AI is not one single technology; it includes different methods designed for different kinds of problems.

AI does not necessarily think or understand in the same way a person does. A system can produce useful results without having human feelings, awareness or common-sense understanding. Its performance depends on its design, data, task and the conditions in which it is used.

Why AI matters

- Automation: handles repeated tasks, such as sorting support requests or filtering spam.
- Pattern recognition: identifies trends or unusual activity that may deserve closer review.
- Information support: helps search, summarise, translate or explain large amounts of content.
- Accessibility: speech-to-text, captions and translation can make information easier to access.

For students, an AI tool can explain a difficult topic in simpler language, make practice questions, give feedback on a draft or help plan revision. The student still needs to understand the material, check important facts and follow course rules.

2. Types of AI

Narrow AI (sometimes called weak AI) is designed for a particular task or a limited group of tasks. Examples include spam filters, recommendation systems, voice assistants and image-recognition tools. Strong performance on one task does not mean the system can perform every intellectual task.

Artificial General Intelligence (AGI) describes a proposed kind of AI that could learn and perform a broad range of intellectual tasks flexibly. There is no universally accepted demonstration of AGI today, and experts may use different definitions.

Superintelligence is a hypothetical concept for an AI that would greatly exceed human capabilities across many domains. It is not a description of typical AI tools currently used by students.

Type	Meaning	Example / status
Narrow AI	Designed for limited tasks	Spam filter; in everyday use
AGI	Broad, flexible ability across many tasks	Concept; not established as achieved
Superintelligence	Hypothetical capability far beyond humans across domains	Speculative concept

3. Machine Learning and Deep Learning

Machine learning (ML) is a branch of AI in which a system learns patterns from data or examples. Developers choose a learning method, train a model and test it on examples that were not used during training. The aim is for the model to perform reasonably well on new inputs, not merely memorise its training examples.

Example: a spam filter can be trained using emails labelled “spam” or “not spam.” It learns statistical patterns that help it classify new messages. It can still make errors when a message is unusual, the data is poor, or people change the way they write spam.

Common approaches to machine learning

- Supervised learning: learns from examples with known labels or answers. Example: predicting house prices from past sales data.
- Unsupervised learning: looks for patterns or groups without supplied answer labels. Example: grouping products with similar features.
- Reinforcement learning: learns which actions tend to earn rewards through interaction with an environment. Example: training a simulated agent to navigate a route.

Deep learning is a type of machine learning that uses neural networks with many layers. These networks can learn complex patterns in images, speech and text. Deep learning supports many modern systems for speech recognition, image analysis, language processing and content generation.

How the terms fit together

ARTIFICIAL INTELLIGENCE
The broad field

↓ includes

MACHINE LEARNING
Learns patterns from data

↓ includes

DEEP LEARNING
Uses neural networks with many layers

Remember: not every AI system uses machine learning, and not every ML system uses deep learning. Some systems rely on explicit rules; others combine rules and learned models.

4. Generative AI, LLMs and Chatbots

Generative AI refers to systems that create content such as text, images, audio, video or code using patterns learned during training. The system generates a new output in response to an input. This does not mean every output is completely original: models can sometimes reproduce memorised material, and generated content may contain errors.

Large language models (LLMs) are trained on large collections of language data. Many learn to predict the next token—a word or part of a word—based on the context that comes before it. By repeating this process, a model can produce paragraphs, answer questions, summarise documents or assist with code.

A high-level view of a chatbot

1. User asks	A student enters a question, such as “Explain supervised learning in simple terms.”
2. Context is prepared	The app may add instructions, conversation context or relevant passages from study material.
3. Model generates	The model generates a response token by token using learned patterns and the context available.
4. App displays	The interface shows the response; some systems also provide retrieved sources or citations.
5. Student checks	The student compares important claims with class notes, textbooks or trusted sources.

What is retrieval-augmented generation (RAG)?

RAG combines information retrieval with text generation. The application searches a chosen collection—such as approved lecture notes, textbook extracts or project documents—and passes relevant passages to the language model. The model uses those passages as context when preparing an answer. This can help connect the answer to the provided material, but it does not guarantee correctness: the system may retrieve an irrelevant passage or misunderstand a useful one.

For an AI Personal Study Assistant, the workflow can be: organise documents into labelled sections, find passages related to the question, generate an answer based on those passages, and display the source section. If the answer is not available in the material, the assistant should clearly say that it could not find the information instead of making up a response.

This is a simplified description of one common design. Different chatbot products may use different models, tools, safety checks and data-handling practices.

5. Applications of AI

Education: explain topics at different levels, create quizzes, translate study material, suggest revision plans and give feedback on writing. AI can support practice, but students should verify facts and follow academic-integrity rules.

Healthcare: support medical-image analysis, organise records, assist research and help clinicians identify patterns. AI output should be reviewed by qualified professionals and is not a substitute for a clinical diagnosis.

Business: forecast demand, detect suspicious transactions, classify customer messages, summarise reports and support inventory planning. Human review is especially important when a mistake could affect a customer or employee.

Daily life: navigation estimates, recommendation systems, voice assistants, spam filtering, photo organisation and accessibility tools such as live captions.

6. Benefits and limitations

- Speed and scale: AI can process large amounts of information quickly, but speed does not guarantee accuracy.
- Personalisation: explanations and practice can be adapted to a learner, although the system may misunderstand the learner's level or needs.
- Consistency: automation can apply the same process repeatedly, but it may also repeat a flawed rule or bias.
- Accessibility: translation and transcription can help more people access information, though quality varies by language and context.

Risks to recognise

Hallucinations: a model may produce a confident but false statement, misunderstand a question or invent a reference. A fluent answer is not proof that it is true.

Bias: training data and design choices can produce unfair or uneven outcomes. Consider whether the system works well for different people and situations.

Privacy: prompts may contain personal or confidential information. Avoid entering sensitive data unless the tool is approved for that purpose and its data practices are understood.

Over-reliance: copying answers without understanding can weaken learning. Attempt the task yourself, then use AI to explain, check or practise.

Misuse: generated material can be misleading or used irresponsibly. Check accuracy, permissions and possible consequences before sharing it.

7. Responsible AI Use

A practical checklist for students

- Verify important claims, dates, statistics and references with reliable sources.
- Open cited sources and confirm that they support the statement; never rely on a citation that cannot be checked.
- Keep passwords, personal identifiers, private student records and confidential project data out of unapproved tools.
- Follow course rules about AI assistance, attribution, collaboration and original work.
- Check whether an answer may be biased, incomplete, outdated or unsuitable for the situation.
- Ask for an explanation and test your understanding instead of only copying the final answer.
- When using AI-generated content, review it carefully and take responsibility for what you submit or share.

8. Comparison: AI, ML, deep learning and generative AI

Term	Core idea	Example	Relationship
AI	Broad field of systems for intelligence-related tasks	Rule-based expert system	Umbrella term
Machine learning	Learns patterns from data	Spam classifier trained on examples	Subset of AI
Deep learning	ML using multi-layer neural networks	Speech-recognition model	Subset of ML
Generative AI	Creates content from learned patterns	Text or image generator	Often uses deep learning; not a synonym for all AI

Practical examples for a study assistant

Example 1 — explain a topic: a student asks for the difference between ML and deep learning. The assistant retrieves the relevant section, explains it in simple language and identifies the section used. The student checks the explanation and attempts a practice question.

Example 2 — missing information: if a question asks for a detail not covered in the uploaded notes, the assistant should say it could not find the answer in the supplied material and suggest checking a textbook or instructor.

Example 3 — revise with feedback: the student answers a quiz question, and the assistant explains why the answer is correct or incorrect, using the study guide as its reference.

9. Glossary of Key Terms

Algorithm	A sequence of steps used to solve a problem or complete a task.
Model	A learned or programmed system that produces an output from an input.
Training data	Examples used to train or fit a machine-learning model.
Inference	Using a trained model to produce a prediction or output.
Neural network	A computing model with connected units that learn patterns through adjustable parameters.
Token	A piece of text processed by a language model; it may be a word, part of a word or punctuation.
Prompt	The instructions or input given to a generative AI system.
LLM	Large language model; a model designed to process and generate language.
Hallucination	A plausible-sounding but inaccurate or unsupported generated claim.
Bias	A systematic skew that can produce unfair or uneven outcomes.
RAG	Retrieval-augmented generation; retrieving source material to inform a generated answer.
Human oversight	People reviewing, guiding or taking responsibility for an AI-assisted process.
Dataset	A collection of examples or records used for analysis or model training.
Evaluation	Testing a system to understand how well it performs and where it fails.

Quick revision notes

• AI is the broad field; ML is a branch of AI; deep learning is a branch of ML.
• Generative AI creates content. Many LLMs generate text by predicting tokens in context.
• A confident chatbot response can still be incorrect or unsupported.
• RAG retrieves relevant passages to help ground an answer, but it cannot guarantee correctness.
• Protect privacy, verify claims, consider bias and follow academic rules.

Memory aid: AI → ML → Deep Learning describes a common subset relationship. Generative AI describes a content-generating capability and often overlaps with deep learning.

10. Self-Test, Answers and Further Reading

Multiple-choice questions — choose one answer for each question.

1. What is AI?
A. Only human-shaped robots B. A field building systems for intelligence-related tasks C. A computer part D. A file format

2. Which is an example of narrow AI?
A. Spam filter B. Proven AGI C. Human brain D. Universal solver

3. Machine learning commonly learns from:
A. Data or examples B. Only handwritten rules C. No input D. Random guesses only

4. Deep learning commonly uses:
A. Multi-layer neural networks B. Paper files only C. One fixed response D. No computation

5. Generative AI is designed to:
A. Only store files B. Create content from learned patterns C. Guarantee truth D. Replace all judgement

6. An LLM primarily models:
A. Language sequences B. Battery charge C. Screen brightness D. Road surfaces only

7. An AI hallucination is:
A. Hardware failure B. A plausible but unsupported or incorrect output C. A verified fact D. A privacy setting

8. RAG is used to:
A. Retrieve relevant source material to inform generation B. Guarantee perfect answers C. Remove all review D. Hide sources

9. Responsible AI use includes:
A. Sharing passwords in prompts B. Trusting every output C. Checking important claims D. Inventing references

10. If the notes lack an answer, a study assistant should:
A. Admit the information was not found B. Guess confidently C. Hide uncertainty D. Stop the student learning

MCQ answer key

1–B · 2–A · 3–A · 4–A · 5–B · 6–A · 7–B · 8–A · 9–C · 10–A

Five short-answer questions with model answers

1. How are AI, ML and deep learning related? AI is the broad field. ML is part of AI that learns patterns from data, and deep learning is part of ML that uses multi-layer neural networks.

2. What is generative AI? Give an example. It creates content based on patterns learned during training. A chatbot drafting a study summary is one example.

3. Why can a chatbot give a wrong answer? It may lack information, misread the context, reflect bias or generate a plausible but unsupported claim. Important facts should be checked.

4. How can RAG help a study assistant? It retrieves relevant passages from approved study material and supplies them to the model. This can ground an answer, though the result still needs checking.

5. Give three rules for responsible AI use. Verify facts, protect sensitive information and follow course rules about AI assistance and original work. Also consider possible bias.

Conclusion

AI and generative AI can make information easier to explore, explain and practise. In education, their value is not just producing a quick answer; it is helping a student understand a concept and become more confident at answering independently. A useful AI Personal Study Assistant should give clear explanations, connect answers to relevant study material, acknowledge when information is missing and encourage verification.

Credible learning resources

- IBM, “What is Artificial Intelligence (AI)?” — https://www.ibm.com/think/topics/artificial-intelligence
- Google for Developers, “Machine Learning Crash Course” — https://developers.google.com/machine-learning/crash-course
- Stanford University, “AI Index Report” — https://aiindex.stanford.edu/report/
- NIST, “AI Risk Management Framework” — https://www.nist.gov/itl/ai-risk-management-framework
- UNESCO, “Guidance for Generative AI in Education and Research” — https://www.unesco.org/en/articles/guidance-generative-ai-education-and-research

These links are starting points for further study. Check the original pages for the latest editions and guidance.